

 VELAMMAL ENGINEERING COLLEGE The Wheel of Knowledge rolls on!	Velammal Engineering College (An Autonomous Institution, Affiliated to Anna University – Chennai) Velammal Newgen Park Ambattur – Red Hills Road , Chennai-600 066
Subject Code :	23PH104T
Subject Title :	PHYSICS FOR ELECTRONICS ENGINEERS

UNIT 1: Electrical Properties of Materials

PART – A

1. Define electrical conductivity.

The electrical conductivity is defined as the quantity of electricity flowing per unit area per unit time maintained at unit potential gradient.

$$\text{Unit : } \Omega^{-1} \text{m}^{-1}$$

$$\sigma = \frac{ne^2\tau}{m}$$

2. Define mobility and drift velocity of electrons.

The magnitude of the drift velocity acquired by the electrons per unit electric field is called mobility.

$$\mu = \frac{v_d}{E}$$

$$\text{Unit : } \text{m}^2 \text{V}^{-1} \text{s}^{-1}$$

Drift velocity is defined as the average velocity acquired by the free electron in a particular direction, due to the application of electric field and is denoted by the letter v_d .

$$v_d = \frac{e\tau_c E}{m}$$

$$\text{Unit : } \text{m} / \text{s}$$

3. Give the postulates of Classical free electron theory.

- The free electrons in the metal moves freely, similar to the gas molecules moving in a vessel and it obeys the classical kinetic theory of gas.
- The forces between the conduction electrons and the ion core is neglected and the total energy of the electron is assumed to be purely kinetic.
- When field is applied the free electron moves in the direction opposite to that of the field direction.
- Due to field applied, they acquire a velocity called drift velocity and the electron velocities in the metal obeys the Maxwell-Boltzmann statistics.

4. Distinguish between relaxation time and collision time.

S.No	Relaxation time	Collision time
1	It is the time taken by the free electron to reach its equilibrium position from its disturbed position, due to the field applied.	It is the average time taken by a free electron between two successive collisions.
2	It is given by $\tau = \frac{l}{v_d}$	It is given by $\tau_c = \frac{\lambda}{v}$

5. Define mean free path.

The average distance travelled between two successive collisions is called mean free path. It is denoted by the letter ' λ '.

$$\lambda = \bar{c}\tau_c$$

Where $\bar{c}$ is the root mean square the velocity of the electron.

6. Define thermal conductivity.

Thermal conductivity of the material is defined as the amount of heat conducted per unit area per unit time maintained at unit temperature gradient.

Unit: $\text{Wm}^{-1}\text{K}^{-1}$

$$K = \frac{nv^2k_B\tau}{2} \quad (\text{or}) \quad K = \frac{Qx}{A(\theta_1 - \theta_2)t}$$

7. Get the microscopic form of Ohm's law and state whether it is true for all temperatures?

According to classical free electron theory current density is

$$J = \sigma E$$

Where σ – electrical conductivity

Since conductivity $(\sigma) = \frac{1}{\rho}$, we can write

$$E = J\rho$$

For a semiconductor of length ' l ' and area of cross section ' A '

$$\text{Resistance} = \frac{\rho l}{A}$$

$$\text{Voltage } V = I \frac{\rho l}{A}$$

(or)

$$V = IR$$

Therefore, microscopically we can write $V = IR$ as $E = J\rho$

Since the resistivity varies with respect to the temperature, the microscopic form of Ohm's law is not true for all the temperatures.

8. What are the difference between drift velocity and thermal velocity of an electron?

S.No	Drift velocity	Thermal velocity
1	Drift velocity is the average velocity acquired by the free electron, in the presence of electric field.	Thermal velocity is the velocity of an electron without any external field.
2	The electrons moving with drift velocity moves in the direction opposite to that of the field direction.	The direction of electrons moving with thermal velocity is random.
3	The velocity is very less, in the order of 0.5 m/s	The velocity is very high, in the order of 10^5 m/s

9. Distinguish between electrical conductivity and thermal conductivity.

S.No	Electrical Conductivity	Thermal Conductivity
1	The co-efficient of electrical conductivity is defined as the quantity of electricity flowing per unit area per unit time maintained at unit potential gradient.	The co-efficient of thermal conductivity is defined as the quantity of heat conducted per unit area per unit time maintained at unit temperature gradient.
2	Electrical conductivity is purely due to number of free electrons.	Thermal conductivity is due to both free electrons and phonons.
3	Conduction of electricity takes place from higher potential end to the lower potential end.	Conduction of heat takes place from hot end to cold end.
4	Unit : $\text{ohm}^{-1} \text{m}^{-1}$	Unit: $\text{Wm}^{-1} \text{K}^{-1}$

10. State Wiedemann –Franz law.

It states that the ratio of thermal conductivity to electrical conductivity of a metal is directly proportional to the absolute temperature.

$$\frac{K}{\sigma} \propto T$$

$$\frac{K}{\sigma} = LT$$

Where L is a constant called as Lorentz number.

11. What is Lorentz number?

The ratio between thermal conductivity of a metal to the product of electrical conductivity of a metal and absolute temperature of the metal is a constant. It is called Lorentz number and it is given by

$$L = \frac{K}{\sigma T}$$

Where, $L = 1.22 \times 10^{-8} \text{ W}\Omega\text{K}^{-2}$ at temperature $T = 293 \text{ K}$.

12. Define Fermi energy level and Fermi energy with its importance.

Fermi energy level: It is the highest reference level of a particle at absolute zero. It is the state at which the probability of electron occupation is 50% at any temperature.

Fermi Energy: It is the maximum energy of the quantum state corresponding to Fermi energy level at absolute zero. It is also the energy of the state at which the probability of electron occupation is 50% at any temperature.

Importance: It is the reference energy level which separates the filled energy levels and vacant energy levels.

13. What are the postulates of Quantum free electron theory?

- The potential energy of an electron is uniform or constant within the metal.
- The electrons have wave nature.
- The allowed energy levels of an electron are quantized.
- The electrons move freely within the metal and they are not allowed to leave the metal due to existence of potential barrier at its surface.
- The free electrons obey Fermi – Dirac statistics.

14. Define zero probability, unit probability and fractional probability.

Zero probability: It is the state which remains always empty and the electrons cannot be filled in it, i.e, $F(E) = 0$

Unit probability: It is the state which is always filled with the electrons, i.e., $F(E) = 1$

Fractional probability: It is the state which may be filled (or) partially filled by electrons (or) it is the state which is filled by the electrons for some time and then will become empty, i.e.,

$F(E)$ will be fraction like 0.5, 0.6 etc.

15. Define density of states and given an example and state its importance.

Density of states $Z(E) dE$ is defined as the number of available energy states per unit volume in an energy interval dE .

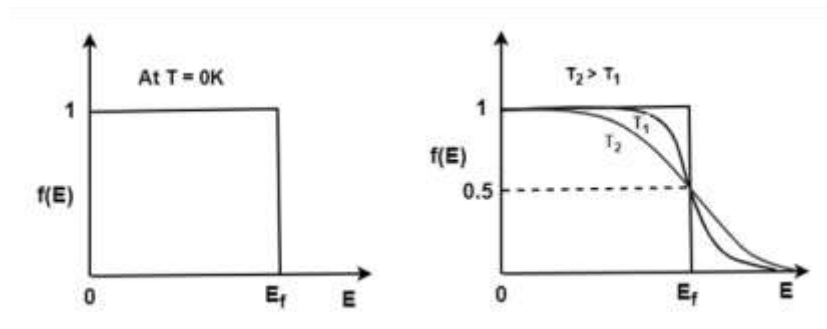
Examples: We can say the density of states of a cubical metal piece

$$Z(E) dE = \frac{\text{Number of available energy states between } E \text{ and } E+dE \text{ in a cubical metal piece}}{\text{Volume of that cubical metal piece}}$$

Importance: It is used to calculate the number of charge carriers per unit volume of any solid.

16. Draw the Fermi distribution curve at 0K and at any temperature TK. (or) How does the Fermi function varies with temperature?

Here at 0 K all the energy states below E_F are filled and all those above it are empty. Now when the temperature is increased, the electron takes an energy $K_B T$ and hence the Fermi function falls to zero.



17. What is meant by effective mass approximation?

For an electron moving in a constant potential field $m^* = m$. But for an electron moving in a periodic potential $m^* \neq m$. Thus, when an electron moves in a periodic potential the free electron mass ' m ' should be replaced by the effective mass m^* and this process is called as effective mass approximation.

18. Write a note on effective mass of an electron moving in a periodic potential.

Effective mass of an electron is the mass of the electron when it is accelerated in a periodic potential and is denoted by m^* .

Explanation: We know mass of an electron is constant, but it is found to vary, when it is accelerated in a periodic potential and this varying mass is called as effective mass, given by

$$m^* = \frac{\hbar^2}{d^2E/dk^2}$$

The effective mass depends on the value of d^2E/dk^2 (ie)

- (i) If d^2E/dk^2 is positive; m^* is also positive
- (ii) If d^2E/dk^2 is negative; m^* is negative
- (iii) If d^2E/dk^2 is zero; m^* becomes infinite

19. Explain the concept of hole and give its advantage.

When the electrons are accelerated in a periodic potential, its mass varies and it moves in the direction opposite to the direction of the applied field. This variation of mass of an electron is called as negative mass behaviour of electrons.

Hole: The electrons with negative mass is called as hole, which has same positive mass as that of an electron but instead of negative charge, the hole will possess positive charge.

Advantage: If we have ' n ' number of empty states in a nearly filled band, then these ' n ' number of empty states can be considered as ' n ' number of holes.

20. Will the effective mass of the electron be negative? Justify your answer.

Yes, the effective mass of the electron can also have a negative value.

Justification: The effective mass of the electron is the mass of the electron when it is accelerated in a periodic potential and is given by

$$m^* = \frac{\hbar^2}{d^2E/dk^2}$$

Here d^2E/dk^2 can vary from positive to negative value, therefore if d^2E/dk^2

is negative, then m^* will also have negative value. Also, this negative effective mass of an electron leads to the concept of hole.

Physically speaking the electrons with negative mass has some positive mass as that of an electron but it has positive charge rather than negative charge.